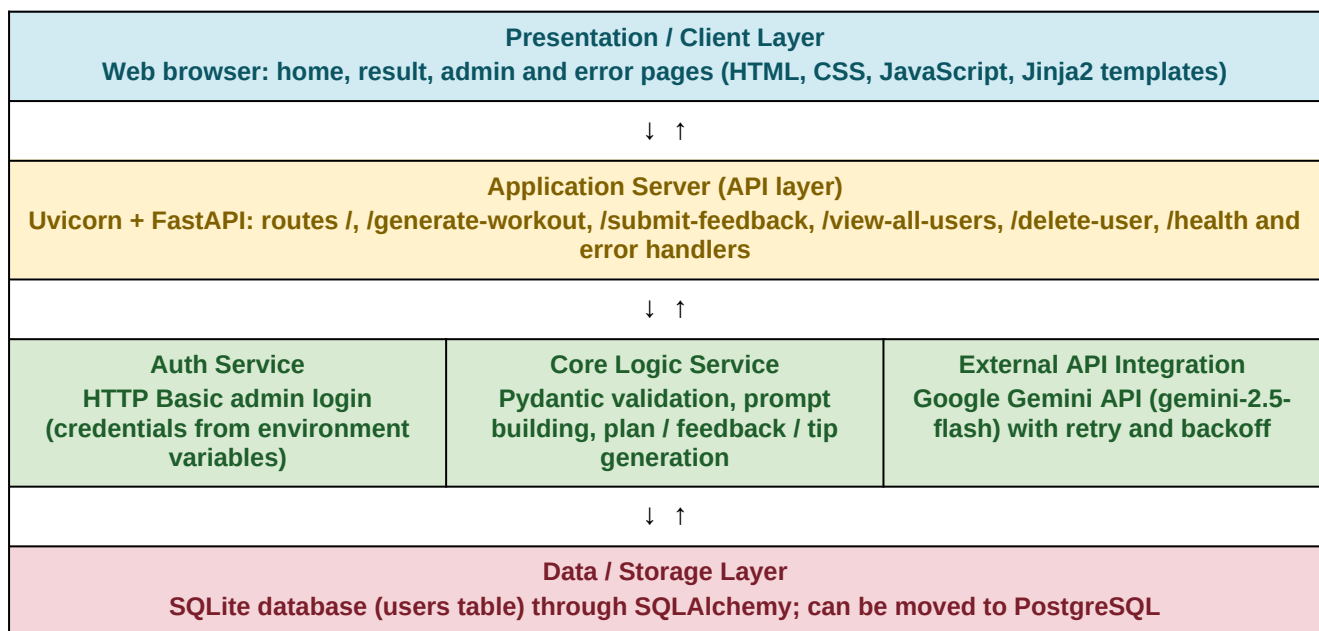


Solution Architecture

Date	29 September 2026
Team ID	SWTID-2026-5841
Project Name	FitBuddy – AI Fitness Plan Generator
Maximum Marks	5 Marks

Solution Architecture Diagram:

High-level view of FitBuddy's Presentation Layer, Application Layer and Data Layer, with the external Google Gemini API.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Shows the input form, the generated plan and tip, the feedback box, the admin list and error pages; JavaScript shows a loading overlay.	HTML, CSS, JavaScript, Jinja2
Application Server	Receives requests, validates input, calls the AI services, reads and writes the database and renders pages; handles errors.	Python, FastAPI, Uvicorn, Pydantic
Core Logic / AI Services	Builds safety-focused prompts, calls Gemini with retry and backoff, returns plan, revised plan and nutrition tip.	google-genai, gemini-2.5-flash
Auth Service	Protects admin functions with HTTP Basic credentials.	FastAPI HTTPBasic, environment variables
Database	Stores users, original plan, updated plan and nutrition tip.	SQLite, SQLAlchemy